Amir Abdullah Zakaria

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Profile

Data Analyst Intern at **The Estée Lauder Companies** and final-year **Computer Science** student at ELTE (CGPA 4.58/5.0) who builds data and AI solutions end to end: **SQL/Python analytics and Power BI dashboards**, **Spark, Kafka and dbt pipelines** over 125M+ rows, and **LLM agents and ML models** served in real time. Automated 16+ processes saving 100+ hours a month.

Achievements

- **2nd Place – Devin Hackathon Budapest, Cognition** (2026)
  Built *preflight-agent* solo, an LLM model-routing cascade for coding agents that fixed every benchmark bug at **32% lower cost** (~74% lower when routed through Devin).
- **Champion – INEC Innovative Engineering Competition**, BME (10/2026)
  Took 1st place across 3 time-boxed industry challenges set and judged by company and university partners.

Professional Experience

Data Analyst Intern, *The Estée Lauder Companies (OneSource, Budapest)* 06/2025 – Present

- Build and maintain **Power BI dashboards** (Savings Tracker, TEM, Temp Labor KPI, Workforce Mgmt) for global finance teams
- Built a star-schema savings model reconciling a **$200M+ vendor baseline**, rolling 294 SAP vendor entities into 146 vendor groups
- Cleansed **10M+ records** in Python and built a **data lake** from 5+ years of contract data with automated refresh
- **Automated 16+ processes** with Python, Power Automate and Office Scripts, saving **100+ hours a month**
- Mapped SAP invoices to procurement towers and POs for director-level reporting; resolved **600+ ServiceNow incidents**

Research Assistant, *Modeling Lab, Dept. of Numerical Analysis, ELTE* 06/2026 – Present

- Benchmark **deep learning** (CNN, FISTA-Net) and optimization methods for ECG signal processing on multiple datasets

Teaching Assistant, *Eötvös Loránd University (ELTE)* 09/2025 – Present

- Lead **Linux, Bash and PowerShell** scripting labs (Computer Systems) and tutor Discrete Mathematics

Projects

github.com/zakaria17amir

Retail Data Platform | *Lakehouse + ML + GenAI: Spark | Kafka | Delta Lake | dbt | Airflow | LightGBM | FastAPI | LangGraph*

- Streamed **CDC** (Debezium → Kafka → Spark Streaming) into a **Delta Lake** lakehouse with **dbt** marts on Airflow (~100K orders)
- **LightGBM** late-delivery model at **2.4× base-rate PR-AUC**, served in 0.5–1.3 s via FastAPI; hybrid **RAG** agents at 0.98 Recall@10

Favorita Stock-out & Promo Analysis | *Big Data + Statistics: Databricks | PySpark | Spark SQL | Unity Catalog | GitHub Actions*

- Built a bronze → silver → gold **medallion pipeline over 42.8M rows**, estimating 8.7M lost units (~2.8% of sales)
- Replaced a naive Poisson test with a negative-binomial model, cutting **false stock-out flags from 21.5% to 4.8%**

Qgate-Agent | *Multi-Agent LLM System: Python | LangGraph | Claude | Kafka | PostgreSQL | Docker*

- Human-in-the-loop quality-gate agents: **0.96 recall**, decisions in 3.6 s (p95) at **$0.0033 each**, zero loss across 138,624 events
- A multi-agent system that decides how many vehicles to quarantine after an end-of-line test failure, with human approval

Education

B.Sc Computer Science, *Eötvös Loránd University (ELTE) · CGPA 4.58 / 5.0 · Stipendium Hungaricum Scholar*

- Thesis: **LEARN-SO**, an AI assistant for natural-language data questions (NLP, text-to-SQL, RAG)

Technical Skills

Programming & Data	Python | SQL | PostgreSQL | Oracle | Statistics
Data Engineering	Apache Spark (PySpark, Structured Streaming) | Databricks | Kafka | Airflow | dbt | Delta Lake
BI & Analytics	Power BI (DAX, Power Query, RLS) | Microsoft Fabric | Excel | SAP | ServiceNow | KPI reporting
AI / ML	LLM agents | LangGraph | RAG | LightGBM | scikit-learn | PyTorch | MLflow
DevOps & Automation	Docker | Terraform | Git | CI/CD (GitHub Actions) | Linux, Bash | Power Automate | Copilot Studio
Languages	English (fluent) | Bengali (native) | Hindi/Urdu (intermediate)